

Topological Energy Condensate Theory: Math 01-12 Summary

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I. CONSOLIDATED MATHEMATICAL STATUS OF TECT PART I

We summarize, at the level of rigor required for submission to *Physical Review Letters/Physical Review X*, the present mathematical status of the Topological Energy Condensate Theory (TECT) in the form in which it appears in the uploaded Part I corpus (Refs. [? ?]) and in the three independent verification codes (`code1--code3`). We distinguish three epistemic tiers: (T1) theorems established without additional assumptions, (T2) theorems established conditionally on explicitly stated hypotheses, and (T3) identified open problems which, if resolved, close the ultraviolet–infrared bridge required for the full emergence programme.

A. Tier T1: Unconditional Results

Theorem I.1 (BCC combinatorial dominance). *For the single-shell isotropic Landau–Brazovskii class (Def. 1 of [?]), the reciprocal star $S_{\text{BCC}}^{\{110\}} \subset \mathbb{R}^3$ satisfies*

$$N_{\text{loop}}^{\text{BCC}} = 6, \quad \mathcal{C}_{\text{BCC}} = \frac{1}{6}, \quad \beta_{\text{MF}}^{\text{BCC}} = \frac{15}{4} = 3.750, \quad (1)$$

and the combinatorial resonance invariant is

$$\Lambda_{\text{BCC}} \equiv \#\{(a, b, c, d) \in S_{\text{BCC}}^4 : \mathbf{k}_a + \mathbf{k}_b + \mathbf{k}_c + \mathbf{k}_d = \mathbf{0}\} = 540. \quad (2)$$

Both values are global maxima over all 12-vector reciprocal stars on S^2 and are exact integers, independent of any coupling or regularization choice.

Theorem I.2 (Selection threshold; Master formula). *Within the 1-loop Landau–Brazovskii effective action, the BCC structure is the unique global minimizer of $\mathcal{F}_{\text{eff}}[\{A_S\}]$ if and only if*

$$u I_{\text{bubble}}(r) > \frac{\beta_{\text{MF}}^{\text{BCC}} - \beta_{\text{MF}}^{\text{FCC}}}{(\mathcal{C}_{\text{BCC}} - \mathcal{C}_{\text{FCC}})/2} = \frac{0.375}{1/48} = 18.0, \quad (3)$$

with the corresponding critical quench depth

$$r_{\text{sel}}^* = \left(\frac{uk_0}{144\pi\sqrt{\kappa}} \right)^{2/3}. \quad (4)$$

[Scope Theorem] The BCC hierarchy $\mathcal{C}_{\text{BCC}} > \mathcal{C}_{\text{FCC}} > \mathcal{C}_{\text{SC}}$ is parametrically stable under: (i) finite shell thickness, (ii) $O(n)$ -symmetric multi-component lifts, (iii) all-loop perturbative bubble dressing, and (iv) continuous parameter sweeps within the isotropic Brazovskii class. No finite-loop correction in the single-momentum-shell scalar theory can invert the BCC/FCC ordering.

B. Numerical Grounding

The three verification codes supply independent empirical support for Theorems I.1–I.2:

1. `code1_free_energy_minimization.py` (analytical free-energy scan): $\lambda_{\text{sel}}^{\text{num}} = 18.00$ within $< 0.05\%$ of Eq. (3).

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2. `code2_pseudospectral_simulation.py` (unbiased pseudo-spectral integration of $\partial_t \Psi = -[r + \kappa(\nabla^2 + k_0^2)]\Psi - (u/6)\Psi^3$ on $N^3 = 64^3$): starting from random noise at $r = -0.5$, $u = 2.0$, the structure factor spontaneously peaks at

$$k_{\text{peak}} = 3.945, \quad k_0 = 3.927, \quad k_{\text{peak}}/k_0 = 1.0046, \quad (5)$$

with angular correlations classified as BCC (twelve $\cos \theta \simeq 1/2$ pair resonances on the critical shell), and the energy density relaxes monotonically to $\langle \mathcal{E} \rangle \simeq -6.3 \times 10^{-2}$.

3. `code3_multishell_resonance.py` (exact integer-lattice counting of $\Lambda(S)$ for BCC, FCC, SC, IQC):

$$\Lambda_{\text{BCC}} = 540, \quad \Lambda_{\text{FCC}} = 216, \quad \Lambda_{\text{SC}} = 90, \quad \Lambda_{\text{IQC}} = 132, \quad (6)$$

with the icosahedral cross-shell peak $\Lambda_{\text{cross}}(\rho=\tau) = 1584$ exactly at $\rho_{\text{lock}}(n=5) = 2 \cos(\pi/5) = \tau$, confirming the Shell-Locking Theorem.

All numerics are consistent with the closed-form predictions to integer precision (code3) or to sub-percent precision (codes 1–2). The reported residuals $m^* = 0.3138$, $h = 0.003$ are within the regime in which the pseudo-spectral ETD1 scheme is uniformly stable.

C. Tier T2: Conditional Results

Proposition I.3 (Transverse $U(1)$ emergence). *Under the three structural hypotheses*

(H1) *gapped volumetric (longitudinal) mode at the condensation scale,*

(H2) *emergent infrared isotropy of the linearized Hessian,*

(H3) *longitudinal/transverse splitting of the quadratic kernel $H_{ij}(\mathbf{k}) = H_L P_{ij}^L + H_T P_{ij}^T + O(a^2 k^4)$,*

the low-energy theory of the BCC condensate reduces to a free Maxwell action $S = -\frac{1}{4} \int F_{\mu\nu} F^{\mu\nu}$ with an emergent $SO(2) \cong U(1)$ transverse-polarization symmetry.

Proposition I.4 (Internal $SU(3)$ and triplication). *Identifying the orientational manifold as $\mathcal{M} = \mathbb{CP}^2 \cong SU(3)/(SU(2) \times U(1))$, one obtains: (i) an emergent Yang–Mills connection $G_\mu \in \mathfrak{su}(3)$, (ii) an integer topological charge $B \in \pi_2(\mathbb{CP}^2) = \mathbb{Z}$ for 3D point defects, and (iii) the exact flavour triplication*

$$N_g = \chi(\mathbb{CP}^2) = 3, \quad (7)$$

enforced by the unbroken discrete cubic symmetry of the residual vacuum.

Proposition I.5 (Derrick-stable fermionic defect). *The next-to-leading gradient expansion of the BCC microscopic kernel $G^{-1}(q)$ generates a Skyrme-type quartic term $E_4 = \kappa_4 \int d^3x [(\partial_i z^\dagger \partial_j z) - (\partial_j z^\dagger \partial_i z)]^2$ with $\kappa_4 \propto a^2 \kappa_2$. Under spatial rescaling $x \rightarrow \lambda x$, $E_2 \mapsto \lambda E_2$, $E_4 \mapsto \lambda^{-1} E_4$, so $\partial_\lambda(E_2 + E_4) = 0$ admits a finite $\lambda^* = \sqrt{\kappa_4/\kappa_2} \propto a$, bypassing Derrick’s theorem at a microscopic, discreteness-fixed core size. A 3D Jackiw–Rebbi zero mode on the hedgehog profile $\Phi(r=0) = 0$, $\Phi(r \rightarrow \infty) = v$ delivers a finite Dirac mass gap.*

Proposition I.6 (Infrared isotropy, one-loop). *In $d = 4 - \epsilon$, the leading cubic-anisotropy operator $\mathcal{O}_{\text{ani}} = \sum_a \varphi_a^4 - \frac{3}{N+2}(\varphi^2)^2$ has RG eigenvalue*

$$\theta_\tau = \epsilon \frac{4 - N_{\text{eff}}}{N_{\text{eff}} + 8}, \quad (8)$$

which is negative whenever $N_{\text{eff}} > 4$. Under the one-loop Wilson–Fisher-type hypothesis, the BCC cubic anisotropy flows to zero in the infrared.

D. Tier T3: Open Problems (Peer-review Exposed Surface)

The following are the precise loci where Part I is *not* yet closed, stated in a form that admits direct attack in Part II:

- (O1) **Derivative-anisotropy irrelevance.** The velocity-mismatch operator has classically marginal scaling; its infrared fate is controlled by the linearized kinetic-flow matrix $\mathcal{M}_Z$ obtained from the Wetterich projection of $\Gamma_k^{(2)}$ onto the three sectoral kinetic terms $(Z_{T,k}, Z_{O,k}, Z_{D,k})$. The statement

$$\text{spec}_{\text{nontriv}}(\mathcal{M}_Z) \subset \mathbb{R}_{<0} \quad (9)$$

is *not yet proved*. This is the single binding obstruction between “infrared isotropy” and “emergent Lorentz invariance.”

- (O2) **Effective component number.** The derivation of $N_{\text{eff}} > 4$ for the full TECT field content (transverse gauge + $\mathbb{CP}^2$ orientational + defect sector) presently relies on a sector-by-sector soft-mode count $N_T + N_{O,\text{soft}} + N_D$ whose value depends on the gauge-fixing and the adiabatic projection. A gauge-invariant derivation is required.
- (O3) **Microscopic matching of $\kappa, u_0, r_{\text{eff}}$.** The mass formula $m \propto \sqrt{|r_{\text{eff}}|\Lambda_{\text{BCC}}/(Ku_0)}$ is dimensionally closed, but the absolute calibration of $\{\kappa, K, u_0\}$ to microscopic BCC lattice parameters is not yet executed. Without this, TECT predicts mass *ratios*, not absolute scales.
- (O4) **Global bundle topology.** The classification of $\mathbb{CP}^2$ defect sectors as $SU(3)$ -bundles over S^3 (monopole charges, Wess–Zumino terms) is sketched but not executed at the level of characteristic classes.
- (O5) **Emergent gravity coupling.** The local sound-speed variation $c_T(x) = \sqrt{\mu(x)/\rho(x)}$ is argued to generate Einstein curvature in the continuum limit, but the matching constant G_{eff} is not yet extracted from the BCC elastic sector.

E. Logical Dependency Graph

$$\underbrace{\Lambda_{\text{BCC}}=540}_{\text{T1, exact}} \longrightarrow \underbrace{\text{BCC ground state}}_{\text{T1, Thm. 2}} \longrightarrow \underbrace{U(1), \mathbb{CP}^2, SU(3)}_{\text{T2, under H1–H3}} \longrightarrow \underbrace{N_g=3, \text{Jackiw–Rebbi}}_{\text{T2}} \longrightarrow \underbrace{\text{IR isotropy}}_{\text{T2, one-loop}} \xrightarrow{\text{O1}} \underbrace{\text{Lorentz}}_{\text{open}}.$$

Everything upstream of (O1) is either proved or conditionally proved; (O1) is the *single vertex* at which the emergence chain currently fails to close. Resolving Eq. (9) converts TECT Part I from a structurally complete programme into a theorem.